EXPERIMENT-3:

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Code:

```
# ============================================

# Decision Tree (ID3 Algorithm)

# Google Colab

# ============================================


import pandas as pd

from google.colab import files

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score


# Upload Dataset

uploaded = files.upload()


# Read CSV File

filename = list(uploaded.keys())[0]

df = pd.read_csv(filename)


print("First 5 Rows:")

print(df.head())


# Remove 'day' column (it is only an ID)

if 'day' in df.columns:

    df = df.drop('day', axis=1)


# Encode categorical columns

encoders = {}
```

```
for col in df.columns:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    encoders[col] = le

# Features and Target
X = df.drop('play', axis=1)
y = df['play']

# Build Decision Tree using ID3 (Entropy)
model = DecisionTreeClassifier(
    criterion='entropy',
    random_state=42
)

# Train Model
model.fit(X, y)

# Predict Training Data
y_pred = model.predict(X)

# Accuracy
print("\nAccuracy:", round(accuracy_score(y, y_pred) * 100, 2), "%")

# New Sample
sample = pd.DataFrame({
    'outlook': ['Sunny'],
    'temp': ['Cool'],
    'humidity': ['High'],
```

```
    'wind': ['Strong']
})
# Encode New Sample
for col in sample.columns:
    sample[col] = encoders[col].transform(sample[col])
# Prediction
prediction = model.predict(sample)
# Decode Prediction
result = encoders['play'].inverse_transform(prediction)
print("\nNew Sample:")
print(sample)
print("\nPredicted Class:")
print(result[0])
```

Output:

```
Choose Files  play_tennis[1].csv
play_tennis[1].csv(text/csv) - 470 bytes, last modified: 8/3/2026 - 100% done
Saving play_tennis[1].csv to play_tennis[1] (2).csv
First 5 Rows:
  day   outlook  temp humidity    wind play
0  D1     Sunny   Hot     High    Weak   No
1  D2     Sunny   Hot     High  Strong   No
2  D3  Overcast   Hot     High    Weak  Yes
3  D4      Rain  Mild     High    Weak  Yes
4  D5      Rain  Cool   Normal    Weak  Yes

Accuracy: 100.0 %

New Sample:
   outlook  temp  humidity  wind
0        2     0         0     0

Predicted Class:
No
```